

Unified Emergency Response Platform for Intelligent Ambulance Matching and Real-Time Emergency Coordination

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Abstract - The rising urban population has placed increasing strain on emergency healthcare systems in India, where traffic congestion, limited ambulance availability, and ineffective communication among the public, ambulance services, and hospitals can delay emergency response. UyirKappan is a unified emergency response platform that combines geospatial computing and real-time communication to coordinate these stakeholders. The system considers spatial proximity, route-based assessment, and ambulance availability rather than relying only on the nearest ambulance. If an ambulance rejects an assignment or fails to respond within the configured time, the request is automatically reassigned to the next eligible ambulance through a cascading fallback mechanism. The platform also provides real-time ambulance location, status, and estimated time of arrival updates, while enabling hospitals to monitor incoming cases and resource availability. This integrated workflow supports coordinated emergency response from request initiation through ambulance dispatch, tracking, hospital preparation, and arrival. The platform therefore provides a unified approach to ambulance selection, dispatch recovery, real-time tracking, and hospital coordination.

Index Terms - Ambulance dispatch, cascading fallback, emergency response, geospatial computing

I. INTRODUCTION

Proper and timely emergency medical response is a key element to better trauma and other time-sensitive emergency outcomes. The Golden Hour is an awareness of the time frame immediately after a serious medical incident and the need to treat the patient correctly during

this crucial time. In the big city though, there are several issues that can impede emergency response, such as traffic jams, limited ambulance services, and poor communication between bystanders, ambulance service and hospital. Geographically closest ambulances may not be fastest due to roadways, route changes, etc.

Today, much of the emergency-response approaches are geared towards specific parts of the emergency chain, such as ambulance routing, vehicle dispatch, hospital management, and bed availability. These solutions enhance specific areas of emergency response, but they are generally viewed in isolation, failing to provide a seamless flow of information between the person reporting the emergency, the ambulance service and the receiving hospital. This underscores the importance of an integrated system to coordinate the entire emergency-response process.

To meet this demand, UyirKappan offers a single platform where bystanders, ambulance drivers, and hospitals can communicate with each other via a shared real-time backend. H3 is used for geospatial indexing to filter out qualified candidates for each ambulance, and Dijkstra is used for route-cost computation to rank the candidates. If the assigned ambulance refuses the assignment, or fails to respond within the time limit set by the configuration, the Cascading Fallback Dispatch mechanism will automatically attempt to match another ambulance. It also offers location and status of ambulance vehicles available in the area, estimated time of arrival, information about incoming vehicles and hospital resources available.

The key functions are to allow emergency stakeholders to coordinate live, to facilitate geospatial and route-based

ambulance allocation, automatically handling ambulance assignments that fail, tracking ambulances in real-time and ensuring hospitals are ready with advance case information and resources visibility. The service is planned to be expanded with traffic-signal pre-emption, in-transit patient telemetry, and the ability to predict the availability of ambulances and ETA using machine learning.

II. RELATED WORK

A. Multi-Sided Emergency Coordination

Rafaqat et al. [1] present EMCON, a digital emergency-response platform to link patients, ambulance providers and hospitals in a low- and middle-income region. Shared resource visibility and real-time coordination between various stakeholders are shown to be of value in their work. UyirKappan adopts a similar multi-sided coordination principle with an extension to the workflow towards automated ambulance matching and on failure to an ambulance assignment, towards recovery.

B. Ambulance Dispatch and Routing

Abdeen et al. [3] introduced a smart ambulance model that takes into account response, travel, and hospital delay times, highlighting the fact that ambulance response cannot be based solely on geographical distance. Becker et al. [7] provide a broader review of dynamic ambulance relocation research, while Schmid [8] considers ambulance dispatch and relocation as time-dependent optimization problems. UyirKappan uses an implementation-based approach involving spatial candidate filtering followed by route-cost ranking and automatic fallback reassignment when the selected ambulance is unavailable.

C. Hospital Selection and Availability

Olivier et al. [5] utilize road network and ambulance travel times to enhance hospital suggestions for EMS. Xu et al. [6] further enhance the decision support of the hospital by integrating the bed availability prediction and routing information. UyirKappan integrates the availability of beds and resources in the hospital coordination process and provides hospitals with information about the cases received. Predictive bed forecasting and formal bed reservation are not currently in the system.

D. Real-Time Monitoring and Estimated Time of Arrival

Mahalakshmi et al. [4] study an Internet of Things (IoT) based architecture for monitoring ambulances for communication between ambulance and hospital. Chu et al. [9] show that other factors besides distance, such as congestion and road conditions, can help improve estimated time of arrival (ETA) estimation. Estimated time of arrival is provided through the application and communication pipeline along with live ambulance location updates. Machine-learning-based ETA prediction is a potential future extension for UyirKappan.

E. Bystander and Community Response

Smith et al. [2] assess the GoodSAM volunteer first responder app, and find a correlation between accepted alerts and increased survival to hospital discharge in OOHCA. GoodSAM mainly focuses on ECI, while in the case of UyirKappan, the bystander application is the means of starting and tracking an emergency request and coordinating with ambulance and hospital services.

F. Research Gap

The studies reviewed show effective practices in areas of emergency response, such as multi-stakeholder coordination, ambulance dispatch, hospital recommendations, real-time monitoring, and community response. These capabilities do not necessarily form a seamless operational process connecting the emergency reporter, ambulance provider, and receiving hospital. UyirKappan solves this integration challenge by implementing a collection of services and features, including geospatial candidate filtering, routing cost-based ambulance ranking, cascading fallback dispatch, hospital availability visibility, and real-time updates in a single emergency-response platform. The main contribution is to integrate these capabilities into an end-to-end workflow and not to develop a new shortest-path algorithm.

III. PROPOSED SYSTEM

A. System Overview

UyirKappan is a unified emergency-response platform that integrates three client facing applications – a Bystander Mobile App, a Driver/Ambulance Mobile App and a Hospital Dashboard with a single back end and a common database. The system views an emergency request as a coordinated process instead of a transactional request/assignment process. Once an emergency is submitted, the backend identifies suitable ambulance candidates through geospatial filtering and route-based ranking, assigns the selected ambulance, and monitors its response. If the assigned ambulance rejects the assignment or fails to respond within the configured time limit, the

The Bystander Mobile Application is the chief means of requesting an emergency. It fetches the user's (requester)

generates a ranked list. This method does not depend only on geographic location because the closest ambulance could not be the most appropriate vehicle if one is available when the road-network conditions and route costs are taken into consideration. Candidate filtering and ranking is done before a Driver is sent an assignment request. This enables the system to choose from a smaller number of "relevant" ambulances and avoids over-calculating routes.

F. Cascading Fallback Dispatch

The system has a Cascading Fallback Dispatch mechanism to deal with the assignment failure. If an ambulance is assigned but rejects the emergency request or fails to respond within the configured timeout period, it is excluded from further attempts for that emergency. The Matching Engine is re-run with the remaining available candidates and the next best ambulance is assigned. This will go on until an ambulance takes the request or no suitable candidate is left. Therefore, if the initial assignment fails the bystander doesn't have to make another emergency request.

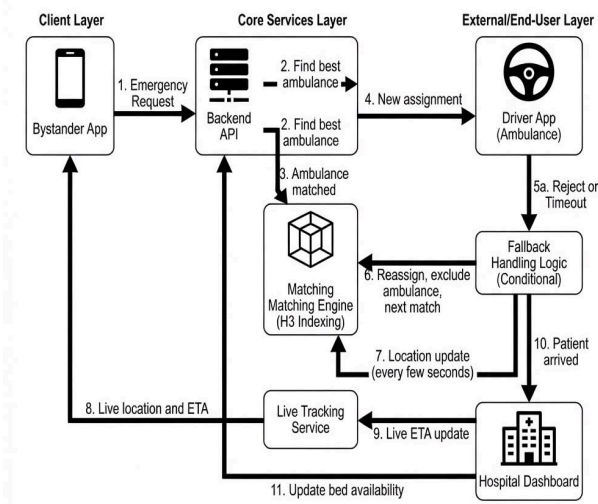


Fig. 3. Driver-side sequence showing ambulance acceptance, rejection or timeout, and cascading reassignment.

G. Hospital Coordination and Bed Availability

The Hospital Dashboard incorporates the receiving hospital into the emergency response process. Case information can be shared with hospital staff for incoming cases of emergency with the estimated arrival time of the assigned ambulance. The hospital staff can update the availability of the bed and resources based on the categories supported by the system. These updates are

recorded in the database and provide the latest available resource information for emergency coordination. This system is not about reserving beds or forecasting hospital capacities automatically, but is about visibility and coordination of the available capacity.

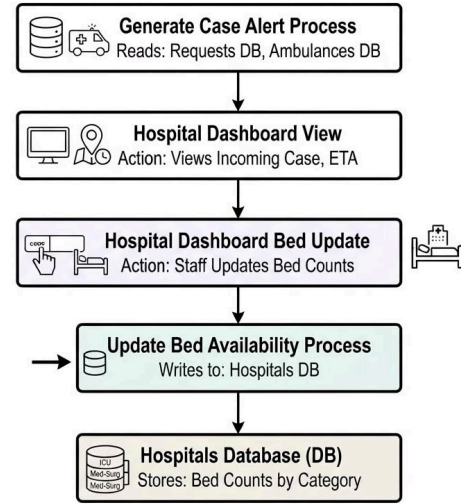


Fig. 4. Hospital dashboard data flow for incoming case alerts and bed-availability updates.

H. Live Tracking and Estimated Time of Arrival

The Driver/Ambulance Application sends its current location to the Backend API periodically, once an Emergency Assignment has been accepted by the Ambulance. The Bystander Application, and the Hospital Dashboard, receives this information from the backend which then communicates with the real-time communication channel to provide the latest ambulance location and estimated time of arrival. This helps all stakeholders involved to have a shared picture of the ambulance's movement throughout the response.

For experimental evaluation involving simulated ambulance movement, the movement data are treated explicitly as simulated data rather than actual live GPS measurements.

I. Hospital Dashboard

The Hospital Dashboard is a web based application for hospital personnel. It has a live feed of incoming cases of emergency, case details and the estimated arrival time of the assigned ambulance. The dashboard also features controls to update bed availability by supported beds (general, intensive care, ventilator beds etc.). These features enable hospital staff to keep up-to-date resource

data and prepare for incoming patients. Updates from the back end enable hospital personnel to track changes without the need for additional communication channels. The dashboard thus serves as the hospital side interface for the coordinated response.

In-transit patient vital-sign telemetry and other high-end monitoring features are expected to be future add-on to the platform.

J. Backend and Database Architecture

The Backend API connects the three client applications to the shared database and offers RESTful services as well as real-time communication for location, status and ETA updates. It accepts and validates emergency requests, maintains request and assignment details, interacts with the Matching Engine, manages ambulance assignments, and broadcasts relevant updates to connected clients. Four main collections exist: Requests for emergency data and time stamps; Ambulances for ambulance location, H3 cell and availability; Request-Attempts for assignment results and for the fallback process such as failure, timeout or rejection; Hospitals for availability of the resources (beds by category).

IV. METHODOLOGY AND IMPLEMENTATION

A. Emergency Processing and System Workflow

UyirKappan's emergency workflow involves a coordinated process that encompasses Bystander Application, backend, Matching Engine, Driver Application and Hospital Dashboard. The back-end validates and logs the emergency when the request is made before initiating ambulance selection. The assigned ambulance accepts or rejects the assignment and then the system continues by tracking and hospital coordination. This workflow maintains a continuous flow of emergency information from request initiation to hospital arrival.

B. Geospatial Matching and Route Optimization

The selection of the ambulances starts with a spatial filtering step that uses H3 to identify the candidates that are geographically relevant. Eligible ambulances are then ranked with travel time as the highest consideration followed by distance, traffic conditions and availability of the ambulances. The road network is used to calculate the route cost using Dijkstra's algorithm, that includes a route taking traffic into account, which is compared to the distance oriented baseline. The route chosen will be used to calculate the estimated time of arrival and is continually updated as the ambulance moves along.

C. Cascading Fallback and Real-Time Coordination

If an ambulance rejects an assignment or does not respond within the configured timeout period, the system records the attempt and excludes that ambulance from further attempts for the same emergency. The Matching Engine is then called again with the remaining eligible ambulances till a candidate is accepted or all the candidates are exhausted. Once accepted, location and status will be sent to all parties, the bystander and driver and the hospital interface, using socket-based communications to keep them synchronized.

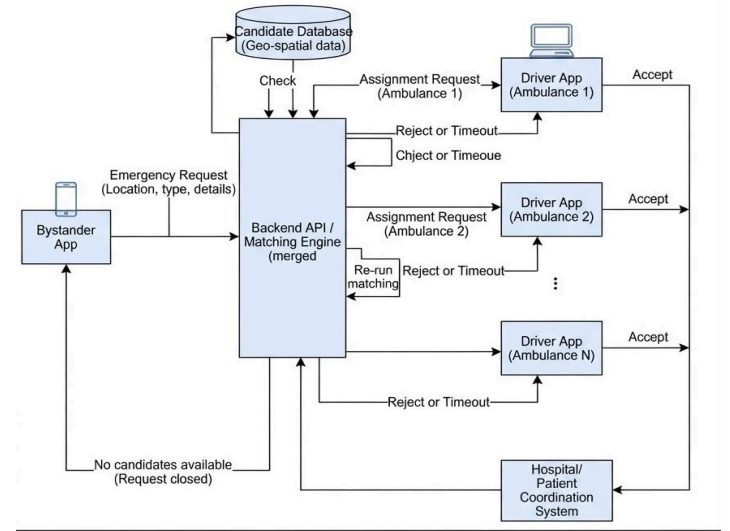


Fig. 5. Emergency request submission, ambulance matching, assignment, fallback, live tracking and hospital coordination sequence of systems.

D. Dynamic Hospital Coordination

Selection of hospitals is embedded in the emergency workflow, and is based on the location of hospitals, resources available at hospitals, emergency-care characteristics, and emergency-response requirements contained in the hospital dataset. Incoming emergency information is passed to the selected hospital, the ambulance information and estimated arrival time. The dashboard provides hospital staff with visibility of the incoming case, ambulance progress, and available resources, supporting preparation before arrival.

E. Backend and Data Management

The back-end offers RESTful services and live communication to facilitate the coordination of emergency requests, assignment of ambulances, hospital information and tracking updates. The system stores

information about requests, ambulances, assignment attempts, hospital and availability of resources in the shared database. The backend is the authoritative state manager, which means that the updates generated by one application will be propagated uniformly to the other connected applications.

V. RESULTS AND ANALYSIS

A. Emergency Coordination

The implemented UyirKappan platform integrates the Bystander Application, Matching Engine, Driver Application, and Hospital Dashboard into a single emergency-response workflow. Emergency requests are matched with ambulances, routed, drivers are assigned, ambulances are tracked, fallback is processed, and hospitals are coordinated. Consistent emergency lifecycle information is maintained across the connected applications through backend state updates.

B. Ambulance and Hospital Selection

The Matching Engine takes into account geographical location, road-based travel attributes, traffic conditions, and ambulance availability, rather than relying only on distance. The decision interface provides visibility into the ambulance selection process by showing the selected candidate and contributing factors. Hospital selection takes into account location, emergency needs, and available resource information, with the selected hospital reflected on the Hospital Dashboard.

C. Route Optimization and Dynamic ETA

The routing component calculates the traffic-aware route using the UyirKappan approach and compares it with the distance-based route used as the reference. The mean dispatch ETA is found to be 7.80 min for the nearest-ambulance baseline and 5.95 min with the multi-factor approach, representing a 23.7% reduction (see Fig. 6).

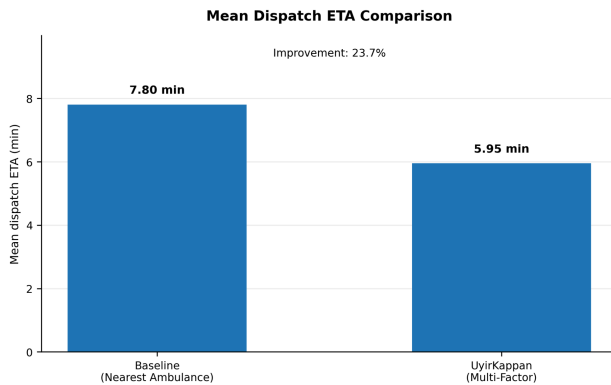


Fig. 6. Mean Dispatch ETA Comparison

There is also an improvement under various traffic conditions, as illustrated in Fig. 7. Under normal traffic conditions, the mean ETA is reduced from 7.83 to 5.99 min, from 7.67 to 5.90 min under moderate traffic conditions, and from 7.94 to 5.86 min under heavy traffic conditions. The location and estimated time of arrival of the ambulance are continuously updated as the ambulance travels along the selected route, providing a live view of the ambulance to the Bystander Application and the Hospital Dashboard.

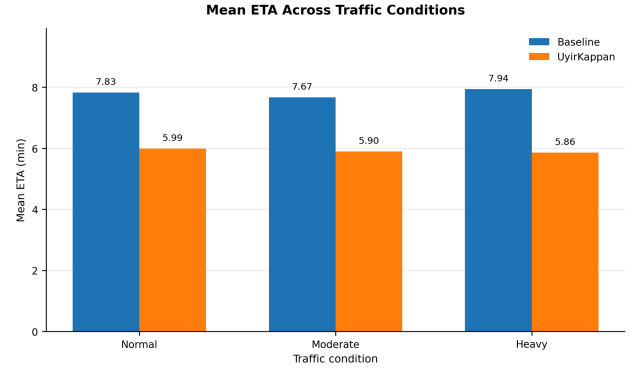


Fig. 7. Mean ETA Across Traffic Conditions

D. Cascading Fallback and Synchronization

If the matching is unsuccessful or the assigned ambulance does not respond within the specified time, the system records the outcome, excludes the candidate, and repeats the matching process. The next eligible ambulance is then considered for assignment without requiring the emergency requester to re-initiate the process. Events such as assignment, fallback, location, and arrival are pushed through the central backend, synchronizing the Bystander Application, Driver Application, and Hospital Dashboard throughout the emergency lifecycle.

E. Overall Analysis

Instead of just showing the final result of ambulance selection, UyirKappan also shows the factors taken into account and the alternative ambulance routes. UyirKappan therefore provides more insight into automated decision-making by showing the factors considered as well as the route alternatives for the ambulance. This assists users in understanding the reason for choosing a specific ambulance or route. Overall, the implementation incorporates a multi-layered approach involving geospatial candidate filtering, ambulance matching, traffic-aware routing, dynamic ETA, cascading fallback, hospital coordination, and real-time tracking within a single emergency-response workflow. The key contribution is the integration of these components to

create a more coordinated decision-support platform rather than the development of a new individual algorithm.

VI. CONCLUSION

UyirKappan has developed a unified emergency-response platform connecting bystanders, ambulance services, and hospitals through real-time coordination. The platform integrates H3-based Ambulance Candidate Filtering, Dijkstra-based Ambulance Routing, Multi-Factor Ambulance Selection, Cascading Fallback Ambulance Dispatch, Hospital Resource Availability and real-time Ambulance Tracking all into one Emergency pipeline. Results for the experimental approach demonstrate that the mean dispatch ETA is decreased from 7.80 min at the nearest-ambulance baseline to 5.95 min using the multi-factor approach, a 23.7% decrease. The system also ensures coordination between the applications, synchronizing the assignment, fallback, location and hospital-related updates on the back end. The main contribution is the assembly of these components into a coordinated decision support platform, not the development of a new algorithm per se. Future plans are to support traffic-signals priority, in-transit patient telemetry, predictive ambulance availability and ETA, better hospital selection and real world validation of the system on a large scale.

VII. REFERENCES

- [1] W. Rafaqat, S. M. A. Abidi, J. Lee, A. A. Javed, and A. I. Mian, "EMCON: A Comprehensive Emergency Response System for Low- and Middle-Income Countries," *Disaster Medicine and Public Health Preparedness*, vol. 19, p. e324, 2025, doi: 10.1017/dmp.2025.10253.
- [2] C. M. Smith, R. Lall, R. T. Fothergill, R. Spaight, and G. D. Perkins, "The Effect of the GoodSAM Volunteer First-Responder App on Survival to Hospital Discharge Following Out-of-Hospital Cardiac Arrest," *European Heart Journal: Acute Cardiovascular Care*, vol. 11, no. 1, pp. 20–31, 2022, doi: 10.1093/ehjacc/zuab103.
- [3] M. A. R. Abdeen, M. H. Ahmed, H. Seliem, T. R. Sheltami, T. M. Alghamdi, and M. El-Nainay, "A Novel Smart Ambulance System—Algorithm Design, Modeling, and Performance Analysis," *IEEE Access*, vol. 10, pp. 42656–42672, 2022, doi: 10.1109/ACCESS.2022.3168736.
- [4] S. Mahalakshmi, T. Ragunthar, N. Veena, S. Sumukha, and P. R. Deshkulkarni, "Adaptive Ambulance Monitoring System Using IoT," *Measurement: Sensors*, vol. 24, Art. no. 100555, 2022, doi: 10.1016/j.measen.2022.100555.
- [5] A. Olivier, M. Adams, S. Mohammadi, A. Smyth, K. Thomson, T. Kepler, *et al.*, "Data Analytics for Improved Closest Hospital Suggestion for EMS Operations in New York City," *Sustainable Cities and Society*, vol. 86, Art. no. 104104, 2022, doi: 10.1016/j.scs.2022.104104.
- [6] Y.-Y. Xu, S.-J. Weng, P.-W. Huang, L.-M. Wang, C.-H. Chen, Y.-T. Tsai, *et al.*, "The Emergency Medical Service Dispatch Recommendation System Using Simulation Based on Bed Availability," *BMC Health Services Research*, vol. 24, Art. no. 1513, 2024, doi: 10.1186/s12913-024-12006-8.
- [7] J. Becker, L. Kurland, E. Höglund, and K. Hugelius, "Dynamic Ambulance Relocation: A Scoping Review," *BMJ Open*, vol. 13, no. 12, Art. no. e073394, 2023, doi: 10.1136/bmjopen-2023-073394.
- [8] V. Schmid, "Solving the Dynamic Ambulance Relocation and Dispatching Problem Using Approximate Dynamic Programming," *European Journal of Operational Research*, vol. 219, no. 3, pp. 611–621, 2012, doi: 10.1016/j.ejor.2011.10.043.
- [9] K.-F. Chu, A. Y. S. Lam, K. H. Tsoi, Z. Huang, and B. P. Y. Loo, "Deep Encoder Cross Network for Estimated Time of Arrival," *IEEE Access*, vol. 11, pp. 76095–76107, 2023, doi: 10.1109/ACCESS.2023.3294345.